NAGA DATHA SAIKIRAN BATTULA

PhD, Data Science

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SUMMARY

PhD researcher in Data Science specializing in human-centered AI, multimodal scientific question answering, and interactive visual analytics. Experienced in developing LLM-driven research prototypes, integrating AI reasoning with human decision-making, and evaluating AI-generated outputs through human and expert feedback.

EXPERIENCE

Doctoral Researcher

NJIT's Intelligible Information Visualization (NiiV) lab

01/2025 – present Newark, NJ

- Designed and developed an interactive Human-AI conversational interface for real-time safeguarding of power grids, integrating anomaly detection, LLM-based interpretations, and interactive visualizations.
- Developed a multimodal attribute-extraction pipeline for data (text and image) analysis to support scientific question answering.
- Developed AI evaluation workflows using human and expert feedback to assess the accuracy, relevance, and usefulness of model-generated analytical responses.

Graduate Research Assistant

NJIT's Intelligible Information Visualization (NiiV) lab

02/2024 – 12/2024 Newark, NJ

- Worked on validation of model-based simulations of Weather Research & Forecast (WRF) Model using alternative physics parameterizations to investigate model behavior and validate simulation outputs.

PUBLICATIONS

Conference Proceedings

- A. Dasgupta, N. D. S. Battula, A. Nakarmi, S. Sen, S. Ghosh, and J. Wenskovitch, "Rationalize: Shared semantic reasoning for human-ai collaboration," in *Proceedings of the Future of Human-Computer Interaction Workshop (FutureHCI '26)*, Association for Computing Machinery, 2026. [Online]. Available: https://par.nsf.gov/biblio/10711054
- A. Nakarmi, N. D. S. Battula, A. Diaz, and A. Dasgupta, "Relevance is not enough: A communication-oriented retrieval system for consequential scientific question answering," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, To appear, Association for Computational Linguistics, 2026. [Online]. Available: https://arxiv.org/abs/2609.06222
- N. D. S. Battula, H. R. Kambhampaty, Y. Vijayalata, and R. N. Ashlin Deepa, "Deep-learning residual network based image analysis for an efficient two-stage recognition of neurological

EDUCATION

Ph.D. Student in Data Science

New Jersey Institute of Technology

Jan 2025 – Expected 2029

GPA: 4.0/4.0

M.S. in Data Science

New Jersey Institute of Technology

Sep 2023 – Dec 2024

GPA: 3.91/4.0
MS Thesis: Surveying the role of visual analytics in human-machine teaming
https://digitalcommons.njit.edu/theses/2946/

B.Tech. in Computer Science and Engineering

Gokaraju Rangaraju Institute of Engineering and Technology

Aug 2019 – May 2023

GPA: 9.53/10

SKILLS

Generative AI

- Architectures & Techniques: Retrieval Augmented Generation (RAG), Agentic AI, LLM Fine-Tuning, Prompt Engineering
- Frameworks & Libraries: HuggingFace Transformers, OpenAI API, LangChain

Human-Centered AI & Research

Human-AI Collaboration, Human-Computer Interaction (HCI), Visual Analytics, Human-Centered AI, AI Evaluation & Benchmarking.

Deep Learning

- Deep Learning Techniques: Reinforcement Learning, Natural Language Processing (NLP)
- Tools: Tensorflow, Pytorch

Programming Languages and Tools

Python (libraries: numpy, pandas, TextBlob, NLTK), C, C++, Java

Databases: Postgres, MongoDB, MySQL

Data Visualization: Tableau, Matplotlib, Seaborn, D3.js

CLOUD & MLOps

disorders," in *2023 2nd International Conference for Innovation in Technology (INOCON)*, 2023, pp. 1–6. DOI: 10.1109/INOCON57975.2023.10101037

- H. R. Kambhampaty, N. D. S. Battula, S. Rentala, S. C. Challa, and A. K, "Network optimization of large-scale cnns for applications in autonomous machines," in *2023 2nd International Conference for Innovation in Technology (INOCON)*, 2023, pp. 1–6. DOI: 10.1109/INOCON57975.2023.10101330
- H. R. Kambhampaty, N. D. S. Battula, Y. Vijayalata, and A. D. R. N, "A novel deep-learning based classification of alzheimer's disease in adults," in *2022 IEEE Delhi Section Conference (DELCON)*, 2022, pp. 1–6. DOI: 10.1109/DELCON54057.2022.9752991

- Cloud Platforms: AWS (EC2, SageMaker, S3, Route 53), Google Cloud (Compute Engine, Vertex AI)
- Infrastructure: Docker, Virtual Machines, CI/CD Pipelines